Integrated Hotel Analytics for Hotel Success, Amenities & Customer Experience in India Report

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COVER PAGE

INTEGRATED HOTEL ANALYTICS FOR HOTEL SUCCESS, AMENITIES & CUSTOMER EXPERIENCE

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DECLARATION

We hereby declare that the project titled **"Integrated Hotel Analytics for Hotel Success, Amenities & Customer Experience"**, submitted as part of the requirements for the course 23CSE452 – Business Analytics, is the work carried out by Team 5 under the guidance of Ms Rohini S.

We confirm that the project work, analysis and results presented in this report are based on our implementation and that all external sources, datasets and references used have been appropriately acknowledged. We also declare that the report has been prepared in accordance with the institution's academic integrity requirements.

ACKNOWLEDGEMENT

We would like to thank our project guide Ms Rohini S for guiding and supporting us throughout this Business Analytics project. We express our gratitude to the faculty members of Amrita Vishwa Vidyapeetham for their opportunity and resources to do this work.

We also extend our gratitude to our team members for their cooperation, contributions and efforts in the data collection, data preprocessing, predictive modelling, text mining, association rule mining, and dashboard development. Last but not least, we recognize the developers and providers of the tools, libraries and data sources used in this project.

ABSTRACT

The hotel industry produces vast amounts of hotel listing data and customer reviews, however, analyzing these data sources separately does not allow for the analysis of hotel performance and customer experience in combination. The aim of this project is to create an integrated hotel analytics framework to investigate the characteristics of hotels, forecast the success of hotels based on their ratings, explore the sentiment of visitors, and uncover the relationships between hotel amenities.

The project employs 29,987 hotel reviews initially collected, real hotel data is consolidated and synthetic data is validated and added to the data set for predictive modelling. Three machine learning algorithms: Logistic Regression, Decision Tree and Random Forest were used to classify hotel success, which is defined as customer rating of 4.0 or higher. The test accuracy of Random Forest was 81.34% and the Macro-F1 score was 0.6571. A total of 26,175 processed reviews from 1,659 hotels were used for text mining and VADER sentiment analysis, with an accuracy of 87.37% in terms of three-class sentiment. The analysis revealed that customer pain points are cleanliness, staff behaviour and maintenance. Moreover, the 1,430 hotel transactions with 33 amenity variables were mined using Apriori association rule mining to generate 154 strong association rules.

The results were merged with an interactive dashboard that included hotel performance and customer feedback and amenity patterns. The framework can be used to inform evidence-based hotel positioning, service enhancement and amenity planning.

EXECUTIVE SUMMARY

Project: Integrated Hotel Analytics for Hotel Success, Amenities & Customer Experience

The hospitality industry has difficulties in understanding how hotels get different ratings from customers, why there are common complaints from guests, and which facilities should be focused on. This project aims to solve these issues using an integrated analytics framework which integrates hotel data, customer reviews and amenity information.

The project was built from hotel listings and hotel reviews from web/API. There were 29,987 review records in the first dataset. Additional synthetic hotel records were validated and added to real ones for predictive modelling. The final solution contains three analytical units:

- **Unit 1 – Data Analysis and EDA:** Explored hotel types, hotel prices, hotel location, hotel rating and amenities to find patterns of hotel success.
- **Unit 2 – Predictive Modelling:** Compared Logistic Regression, Decision Tree and Random Forest to classify hotels based on a rating threshold of 4.0.
- **Unit 3 – Advanced Analytics:** Customer opinions and repeated complaints were identified with VADER sentiment analysis and combinations of amenities were found that are often mentioned together in the data using the Apriori association rule mining.

Key project outcomes

Performance indicator	Achieved result
Random Forest prediction accuracy	81.34%
Random Forest Macro-F1	0.6571
Three-class sentiment accuracy	87.37%
Reviews processed	26,175
Strong amenity association rules	154
Hotels included in amenity analysis	1,430

Key business findings

Poorly maintained facilities and staff behaviour were the next most common negative customer concerns, behind cleanliness and hygiene. Positive customer experiences related to location and food were also identified. Amenity analysis found combinations like Business Center + Pool -> Spa with a confidence of 74.02% and a lift of 6.190.

Business value and recommended actions

The project offers a single source of information on hotel performance, customer feedback and amenity patterns to hotel stakeholders. The results can be used to prioritize housekeeping and maintenance, enhance staff service, measure value for money and assess the value of complementary amenity provision before an investment decision.

The final interactive dashboard combines the results from all three analytical units, enabling users to delve into the hotel characteristics, predictive results, customer sentiment and associations with amenities.

The results should be viewed in the light of the limitations of the project, including the use of synthetic records for predictive modelling, the limited classification of neutral sentiment and the lack of real amenity usage and hotel revenues data. The framework could be further validated with independently collected real-world hotel data.

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LIST OF ABBREVIATIONS

Abbreviation	Full form
BA	Business Analytics

API	Application Programming Interface
EDA	Exploratory Data Analysis
NLP	Natural Language Processing
VADER	Valence Aware Dictionary and sEntiment Reasoner
PCA	Principal Component Analysis
RF	Random Forest
LR	Logistic Regression
DT	Decision Tree
ARM	Association Rule Mining
KPI	Key Performance Indicator
CV	Cross-Validation
ROC-AUC	Receiver Operating Characteristic – Area Under the Curve
F1	F1-Score
HTML	HyperText Markup Language
CSV	Comma-Separated Values
TF	Term Frequency
ID	Identifier
CTGAN	Conditional Tabular Generative Adversarial Network
SDV	Synthetic Data Vault

1. Introduction, Business Objectives and Literature Review

1.1 Context and scope

The hotel industry is very competitive and the performance of a hotel depends on pricing, location, hotel category, amenities, service quality and customer satisfaction. These factors can be extracted from online hotel listings and customer reviews, but doing so is difficult as there is a lot of structured and unstructured data. Hotel managers must be familiar with the factors that lead to positive customer reviews, pinpoint areas of concern, and uncover combinations of amenities that can help them make sound business decisions.

The project aims to study the characteristics of hotels, their prices, ratings, geographical differences and amenities to find patterns that correlate with the success of the hotel. It builds and compares Logistic Regression, Decision Tree and Random Forest models to predict hotel success based on ratings, implements NLP and VADER sentiment analysis to detect customer satisfaction and common complaints, and uses the Apriori algorithm to find the most common combinations of amenities. The key analytical questions are: What are the relationships between hotel attributes and customer ratings; what are the patterns of success for various hotel attributes; what are the common complaints of customers and what are the common occurrences of various hotel attributes.

The scope includes a list of hotels in India and their customer reviews scraped from the web and provided by APIs. It covers exploratory analysis at the level of the hotel, predictive modelling, customer review analysis and discovery of association of amenities in an interactive dashboard with all analytical outputs. The success of a hotel is determined by the 4.0 or higher rating given by the customer and not by profit and loss. The dataset contains both real and synthetic hotel records and the analysis is not based on measuring the revenue of the hotels or their occupancy or return on investment. Also, association rules only show co-occurrence and not causation.

The general goal is to build a data-driven hotel analytics system that will be able to link hotel features, predictive analysis, customer feedback and amenity usage with the purpose of providing actionable recommendations to enhance hotel performance and customer satisfaction.

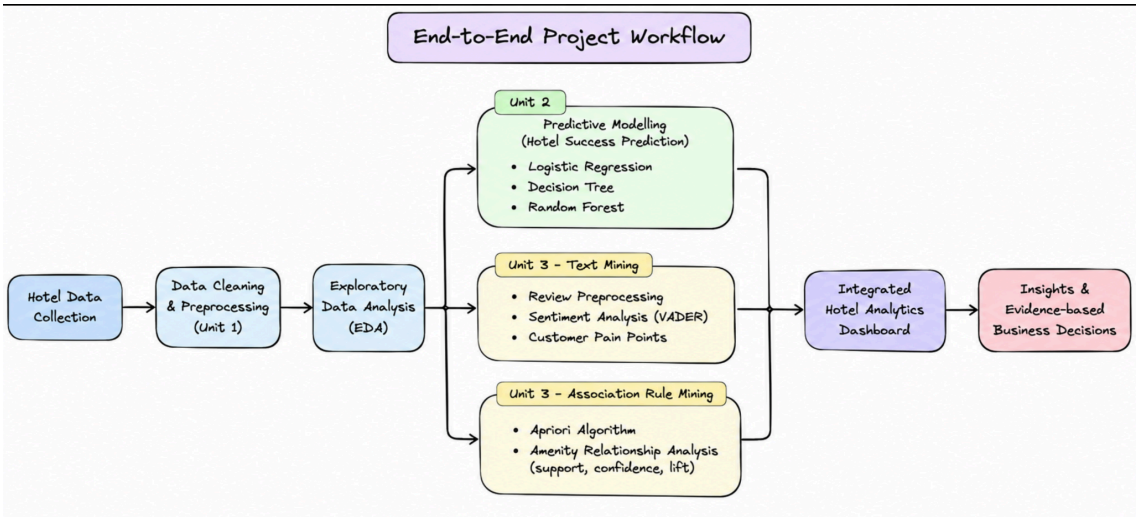


Fig 1. Overall Project Workflow

1.2 Related analytics work and research gap

Existing research in hospitality analytics has explored hotel amenities, customer reviews and sentiment analysis. The following studies provide the foundation for the proposed project.

Research Paper	Dataset	Methodology	Key Findings/Limitations
Muslim et al. (2024) – Machine Learning Models for Hotel Review: A Case Study of Hotels in Perak, Malaysia[1]	2,993 Google hotel reviews from Perak, Malaysia	Naïve Bayes, Support Vector Machine (SVM) and Random Forest for sentiment classification	Naïve Bayes achieved 76% accuracy, SVM 70% and Random Forest 74%. The study focuses on review sentiment classification and reports difficulty handling neutral sentiments.
Comprehensive Examination of Online Reviews Divergence Over Time and Platform Types (2024)[2]	75,000 hotel reviews collected from four online platforms	Comparative analysis of review scores, sentiment, topics	The study focuses on cross-platform review analysis rather than integrating hotel success prediction and amenity association mining.
Le et al. (2026) – Unlocking Insights into Customer Sentiment Analysis[5]	Online hotel ratings and customer sentiment	Customer sentiment analysis and loyalty-based analysis	Examines how customer loyalty relates to online hotel ratings, providing a basis for understanding customer sentiment and hotel evaluation.

Table 1. Comparison of Existing Research

Previous research has analyzed the sentiment of hotel reviews and customer experience on specific platforms, but most of the previous research focuses on certain aspects of hotel performance. A common framework that links hotel features, hotel success prediction, customer feedback analysis, and hotel amenity relationships is possible. The proposed project is designed to fill this gap and involves the integration of exploratory data analysis, machine learning for predicting hotel success, sentiment analysis and association rule mining. It also combines the results into an interactive dashboard, offering a unified perspective on hotel performance and customer satisfaction.

Business objective	KPI / unit	Baseline / target	Relevant unit or method
Predict hotel success	Macro-F1 score	Baseline: 0.4495; Target: ≥ 0.65	Unit 2 – Random Forest
Analyze customer experience	Processed reviews	Target: 26,175 reviews	Unit 3 – Text Mining
Discover amenity relationships	Support and lift	Minimum support: 5%; lift greater than 1	Unit 3 – Apriori

Table 2. Business Objective , KPI , Baseline and Target

2. Dataset Collection, Validation and Preparation

2.1 Provenance, coverage and challenges

2.1.1 Data Source and Collection Method

The dataset was gathered using an original dataset of hotels that have been collected using the help of the Internet via the SerpApi tool in order to gather information from Google Hotels and Google Hotels Reviews. This data includes hotel names, their location, category, price range, rating, number of reviews, amenities and customer reviews text.

The dataset was not taken from pre-existing databases from Kaggle, UCI and GitHub websites. The original dataset consisted of 29,987 review records, 19 original variables and 1,685 unique real hotels.

The time period of data gathering is stated as 30 September to 2 October without the year being stated explicitly in the documents.

2.1.2 Dataset Coverage and Structure

Raw data had 29,987 observations and 19 features, which were increased to 42 features after data cleaning and feature engineering. In review 1, a total of 1,685 actual hotel reviews and 2,887 simulated hotel reviews were utilized. After the elimination of 21 hotels having no rating values or invalid rating

values, 4,551 hotel reviews were left for modelling purpose. In review 2, a total of 26,175 reviews for 1,659 hotels were utilized for text mining purposes and a separate dataset of 1,430 hotels along with 33 binary amenity features was utilized for ARM analysis. Data has categorical, binary and text type features where analysis is performed at review/hotel level. Success in review 1 is defined as the rating greater than or equal to 4.0 as 1.

Hotel name	Location	Price (₹)	Rating	Review text	Review date	Source	Amenity count
FabHotel Gita Residency	Visakhapatnam	1,256	5.0	Rooms are very spacious and cleaning is good	A month ago	Google	7
FabHotel Gita Residency	Visakhapatnam	1,256	5.0	Good rooms	A month ago	Google	7

Table 3.: Sample Records from the Hotel Review Dataset

2.1.3 Data Collection Challenges and Mitigation

Data preprocessing entailed imputing or omitting missing price, rating and amenity values, standardizing rating scales to a five-point scale and renaming amenity descriptions. Missing review content was identified but relative dates remained text while duplicates were identified only if they could not be identified otherwise, and hotels with missing amenity data were dropped from the ARM analysis. Reviews were provided by Google to the extent of 95.1%.

2.1.4 Synthetic Data Generation and Validation

The synthetic hotels' data was generated by CTGAN[7], an algorithm from the SDV library [3], based on 1,685 hotels for 300 epochs and a batch size of 500 and random seed fixed to 0. Among 3,000 synthetic hotels, 2,887 passed the range, binary values, derived variables and duplicates check. The median value of the rating of real hotels is 4.20 with a 70.8% success rate, while it is 4.86 and 88.0%, respectively, for synthetic hotels. That means there was some disparity between the actual distribution of data and the synthetic one. Since the success variable was part of the training process, there may be some reproduction of existing patterns in synthetic records.

2.2 Data Quality, Privacy and Ethics

2.2.1 Data Quality Checks and Class Imbalance

As a result of the data quality check, there were 2,713 missing prices, 4,138 missing amenities and 3,706 missing reviews. There were no duplicate records; however, there were 129 potentially duplicated records among 58 clusters. Reviews' ratings have been scaled down to a five point scale, price outliers have been determined based on the interquartile range and 21 hotels that had either missing or incorrect target ratings have been eliminated from the Review 1 dataset. There were 3,647 successful hotels (79.75%) and 925 unsuccessful hotels (20.25%).

2.2.2 Privacy, Ethics and Data Protection

Personal names and identities of customers were not considered while analyzing the data. The data was analyzed for academic research purposes with access restricted to only the team members. Nevertheless, formal permissions could not be acquired since there is no documentation of consent and data sharing permission.

2.2.3 Potential Biases and Limitations

There is the possibility of source and selection bias due to the 95.1% of reviews that are sourced from Google, and hotels that have more online presence might be over-represented. Lack of information such as price, facilities, and reviews will influence the study, since the ratings provided by the customers are the only ones available. The synthetic hotels had higher median ratings and success rate compared to the real ones.

2.3 Transformations and Usable Dataset

2.3.1 Data Transformations and Preprocessing

Cleaning of the dataset involved duplicate checking, standardization of rating values to 5, imputation of missing data, and identifying price outliers using IQR method. New variables including log_price and amenity_count were created, while aggregation of hotel features was based on hotel_id. In terms of the first review, amenity variables were transformed to binary form, categorical variables were one-hot encoded, numeric variables were standardized and PCA was conducted. Stratified splitting was performed with an 80:20 ratio, while preprocessing was done on the training set only to avoid data leakage. The variable customer_rating was not included in predictors since it was part of the target definition. In the

second review, preparation of review text for sentiment analysis and n-grams was conducted, while binary amenity variables were needed for ARM.

2.3.2 Before-and-After Dataset Counts

There were 29,987 records in the initial dataset, which was the same number of cleaned records. The first review yielded 4,572 records of hotels that were reduced to 4,551 for modelling purposes, where 21 records were not used. The second review had 26,175 records for text mining and 1,430 hotels with 33 amenity attributes for ARM.

2.3.3 Mapping of Final Dataset Variables to Project Units

Variable	Type / meaning	Origin / preparation	Used in
hotel_id	String – hotel identifier	Original data; used for grouping	U1, U2, U3 M1, U3 M2
location	Categorical – hotel location	Standardised	U1, U2, U3 M1
hotel_category	Categorical – hotel classification	Cleaned	U1, U2, U3 M1
price, log_price	Numeric – nightly price	Original price and log transformation	U1, U2
customer_rating	Numeric – hotel rating	Normalised to five-point scale	U1, U2
success	Binary – prediction target	Derived using rating threshold of 4.0	U1, U2
review_text	Text – customer feedback	Original review content	U3 M1
text_light	Text – lightly cleaned reviews	Text preprocessing	U3 M1
text_processed	Text – processed reviews	Text preprocessing for n-grams	U3 M1
vader_compound	Numeric – sentiment score	VADER analysis	U3 M1

amenities	Text/list – hotel facilities	Standardised and grouped by hotel	U1, U3 M2
amenity_count	Integer – number of amenities	Feature engineering	U1, U2
amenity_*	Binary – individual amenities	Converted into 0/1 indicators	U1, U2, U3 M2
PCA1, PCA2	Numeric – principal components	PCA on amenity indicators	U1
data_type	Categorical – Real or Synthetic	Assigned during dataset preparation	U1, U2

Table 4: Mapping of Final Dataset Variables to Project Units

U1 = Unit 1 (Data Preparation and EDA); U2 = Unit 2 (Predictive Modelling); U3 M1 = Unit 3, Method 1 (Text Mining); U3 M2 = Unit 3, Method 2 (Association Rule Mining).

3. Exploratory Data Analysis and Visualization (Unit 1)

Exploratory Data Analysis (EDA) examined the 4,551 cleaned hotel records to evaluate target class balance, real-versus-synthetic distribution differences, pricing behaviour, regional and category variations, and amenity structures linked to customer satisfaction.

3.1 Target Distribution and Source Composition

Hotel success is defined as a customer rating ≥ 4.0 (class 1). Across the 4,551 properties, 3,718 (81.70%) are successful and 833 (18.30%) are unsuccessful, indicating a clear positive class skew. Comparing data sources highlights a distributional shift: real hotels (n = 1,664) have a median rating of 4.20 (mean 4.13 ± 0.52) and a 70.79% success rate (1,178 successful; 486 unsuccessful), whereas CTGAN synthetic hotels (n = 2,887) show a higher median of 4.86 (mean 4.63 ± 0.52) and an 87.98% success rate (2,540 successful; 347 unsuccessful). Both cohorts retain negative-class examples, confirming class diversity without collapse (Figure 2).

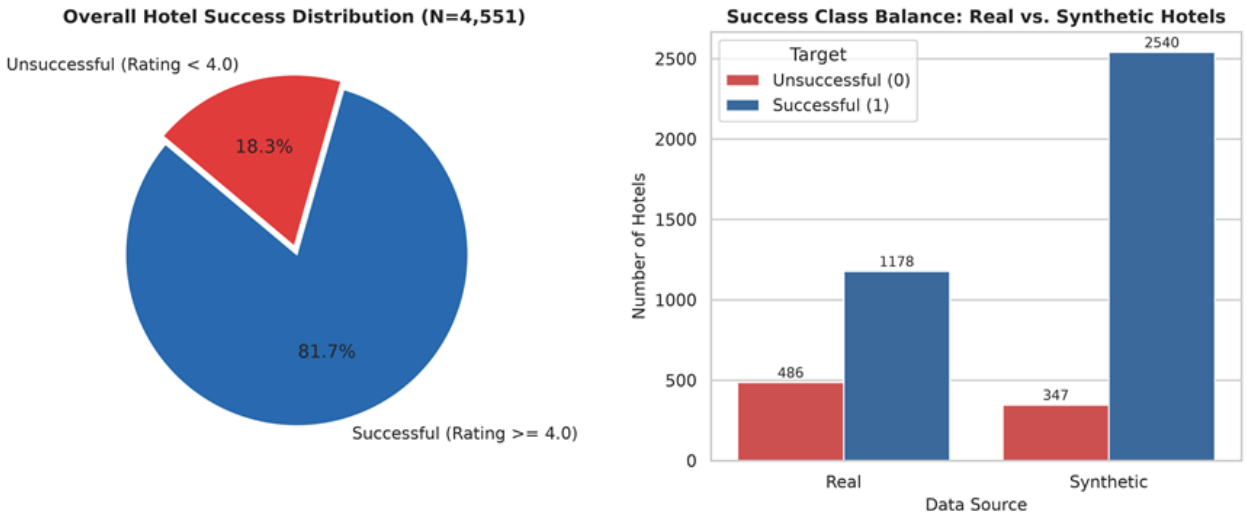


Figure 2. Overall Hotel Success Distribution and Source Balance (Real vs. Synthetic).
[Tool: Python Matplotlib/Seaborn | Source: combined_real_synthetic_hotels.csv / Cell 25]

3.2 Pricing Dynamics and Correlations

Room rates are strongly right-skewed (₹392 to ₹61,327; median ₹2,650; imputed mean ₹4,077), motivating a logarithmic transformation (log_price) for modelling. Linear correlations with customer ratings are weak across the combined data: room price correlates at r = 0.086 (n = 4,362 raw non-null prices; r = 0.182 real-only), while amenity count correlates at r = −0.134 (r = 0.112 real-only). Higher prices or more amenities alone do not guarantee higher guest ratings.

3.3 Category, Location and Amenity Differentials

Success rates scale consistently with hotel category: 1-star (69.7%), 2-star (74.4%), 3-star (79.1%), 4-star (89.0%), and 5-star (92.0%), alongside Uncategorized/Unknown listings (83.6%). Across Indian cities, Goa achieves the highest success rate (88.6%, n = 466) and Hyderabad the lowest (72.0%, n = 353), while major hubs such as Bengaluru (83.6%), Chennai (81.6%), Mumbai (79.6%), and Delhi (75.0%) fall in between. Among the 15 model-ready amenity indicators, Bar (+11.6 percentage points), Free Parking (+9.1 pp), Accessibility (+8.2 pp), Free Wi-Fi (+7.7 pp), and Parking (+7.5 pp) show the widest descriptive success-rate differentials (Figure 3).

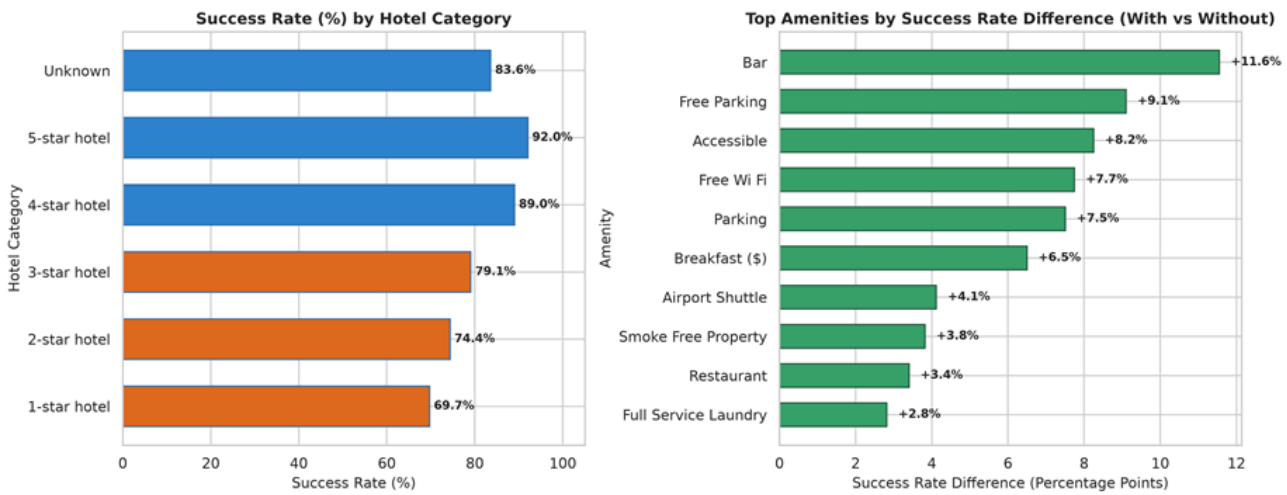


Figure 3. Hotel Success Rate (%) by Category Tier and Top Amenities by Success Rate Differential.
[Tool: Python Matplotlib/Seaborn | Source: clean_hotels_df / Cells 31 & 34]

3.4 PCA Structure and Modelling Implications

Principal Component Analysis (PCA) across the 15 standardized amenity flags captures 48.25% of total variance in two components (PC1: 40.38% core services such as restaurant, parking, and Wi-Fi; PC2: 7.87% dining/accessibility features such as in-room kitchen and breakfast). Key EDA takeaways guiding Unit 2 predictive modelling are:

- **Non-linear interactions:** Weak linear correlations favour tree-based ensembles (Random Forest) over purely linear classifiers.
- **Feature set:** Combining log_price, hotel category, city, and individual amenity indicators captures key performance variance.
- **Imbalance handling:** The 81.7% majority class skew requires stratified 80/20 splits and class-balanced weighting.

4. Feature Engineering and Predictive Modelling (Unit 2)

4.1 Prediction Target and Feature Selection

Unit 2 focuses on predicting hotel success using a binary classification approach. Hotel success was defined using customer ratings: hotels with a rating of 4.0 or above were classified as successful (1), while hotels below 4.0 were classified as not successful (0).[8]

The final model-ready dataset contained 4,551 hotel records and 21 predictor features. The selected predictors included hotel price, log-transformed price, review count, amenity count, individual amenity indicators, hotel category and location. Customer rating and the success target were excluded from the predictor variables to prevent target leakage.

Feature engineering included:

- Log transformation: Applied to hotel price to reduce the influence of extreme values.
- Amenity count: Created to represent the total number of amenities available at each hotel.
- One-hot encoding: Applied to categorical variables such as hotel category and location.
- PCA: Applied to 15 standardized amenity indicators, reducing them to two principal components that explained 48.25% of the total variance. This was used for visualization and exploratory analysis of amenity patterns.

4.2 Predictive Models and Validation

Three classification algorithms were selected to compare different modelling approaches:

- Logistic Regression: Used as a baseline linear classification model.
- Decision Tree: Selected for its interpretable, rule-based predictions.
- Random Forest: Used as an ensemble model to capture complex and non-linear relationships.

The dataset was divided using a stratified 80:20 train-test split, resulting in 3,640 training records and 911 testing records. Stratification maintained the class distribution across both sets. Five-fold cross-validation was used during hyperparameter tuning, with Macro-F1 selected as the optimization metric because of the imbalanced target distribution.

Preprocessing was implemented using Scikit-learn pipelines and ColumnTransformer:

- Numerical features: median imputation and StandardScaler.
- Categorical features: most-frequent imputation and One-Hot Encoding.

Hyperparameter tuning used the following configurations:

Model	Tuned Hyperparameters
Logistic Regression	C = 10; class_weight = balanced

Decision Tree	max_depth = 7; min_samples_split = 10
Random Forest	300 trees; max_depth = 10; min_samples_split = 5; class_weight = balanced

Table 5. Hyperparameter Table

4.3 Model Evaluation and Comparison

All three tuned models were evaluated on the same held-out test dataset. Accuracy, Balanced Accuracy, Macro-Precision, Macro-Recall, Macro-F1 and Weighted-F1 were considered.

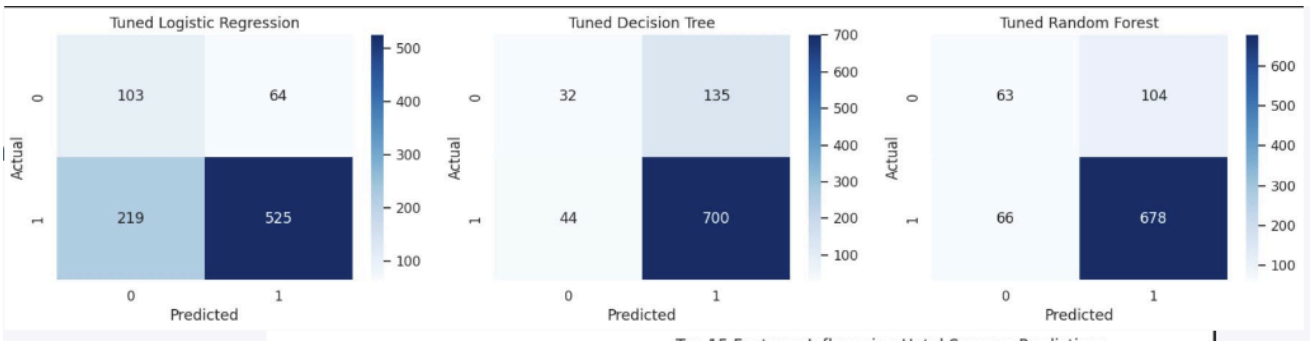


Fig 4. Confusion Matrix of Predictive Models

Model	Validation scheme	Test metrics	Interpretability / bias evidence
Baseline	Stratified 80:20 split	Accuracy: 81.67%; Macro-F1: 0.4495; Balanced Accuracy: 50.00%	Majority-class prediction; poor minority-class recognition.
Tuned Logistic Regression	5-fold CV; held-out test	Accuracy: 68.94%; Macro-F1: 0.6045; Balanced Accuracy: 66.12%	Linear model; provides a comparison against the tree-based approaches.
Tuned Decision Tree	5-fold CV; held-out test	Accuracy: 80.35%; Macro-F1: 0.5750; Balanced Accuracy: 56.62%	Provides interpretable decision rules but has weaker class-balanced performance.
Tuned Random Forest	5-fold CV; held-out test	Accuracy: 81.34%; Macro-F1: 0.6571; Balanced Accuracy: 64.42%	Feature importance identifies influential hotel attributes and amenities.

Table 6. Predictive Model Comparison

The tuned Random Forest was selected as the final model because it achieved the highest cross-validation Macro-F1 (0.6693) and the highest test Macro-F1 and Weighted-F1 among the three tuned models. It also achieved 81.34% test accuracy.

4.4 Interpretability, Leakage Audit and Limitations

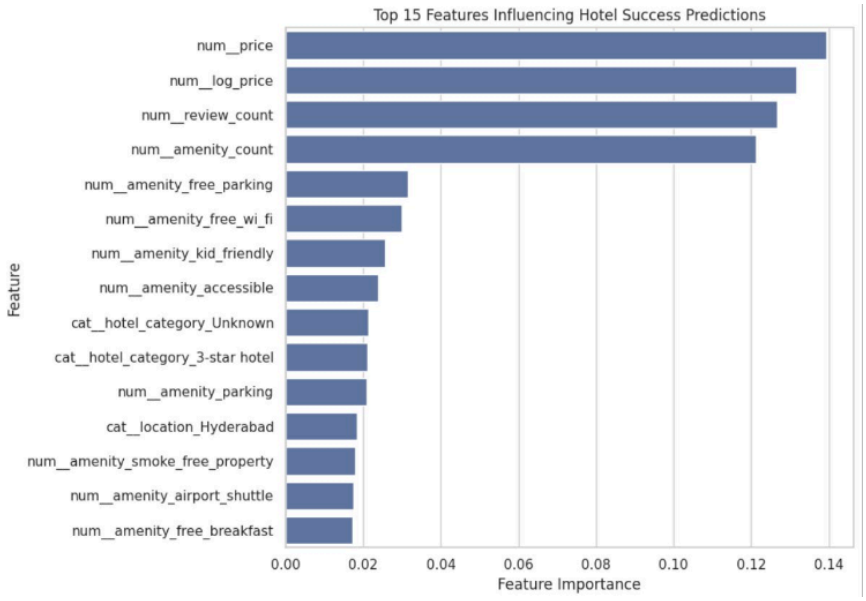


Fig 5. Top Feature Importance

The following checks were taken into consideration when developing the model:

- Customer rating and the success target were not included in the predictor matrix.
- To avoid any information from the test data being used to affect the training, pipelines were developed that included preprocessing steps.
- There were no duplicate hotel IDs or rows in the consolidated data.
- The target was unbalanced with 3,718 successful hotels (81.7%) and 833 unsuccessful hotels (18.3%). Therefore, the following metrics were used in addition to accuracy: Macro-F1 and Balanced Accuracy.

The model yields an estimate of the success of the hotel based on the rating, and the available hotel characteristics. The results of its modelling should be viewed with caution, as the modelling data set contains synthetic data, and the analysis does not provide causal relationships. The results should be further validated with independent real-world hotel data.

5. Advanced Analytics (Unit 3): Two Selected Methods

5.1 Method 1 — Text Minting

The text mining and sentiment analysis were used to gain insight into the customers' experiences, to see how satisfied they were with the hotel and to find out what the most common complaints were from feedback on the hotel. The first dataset consisted of 29,987 reviews for 1,659 hotels, with 26,175 reviews being kept after the preprocessing.

HTML tags, urls, translation artefacts and unwanted characters were removed from the reviews. Negations were kept to preserve meaning of customer feedback and tokenisation, stop-word removal and lemmatisation were performed. Themes of the hotel experiences were identified using word-frequency analysis and bigram and trigram extraction.

Sentiment analysis was performed by using VADER (Valence Aware Dictionary and sEntiment Reasoner) in Python and NLTK [4]. It assigns a compound score between -1 and +1 and categorizes the reviews as negative (≤ -0.05), positive (≥ +0.05) or neutral (between -0.05 and +0.05). The predictions were compared with reference labels based on ratings.

The sentiment analysis has 87.37% accuracy, 0.6356 Macro-F1 and 0.9501 ROC-AUC. Out of 26,175 reviews, 76.3% were positive, 18.6% negative and 5.1% neutral.

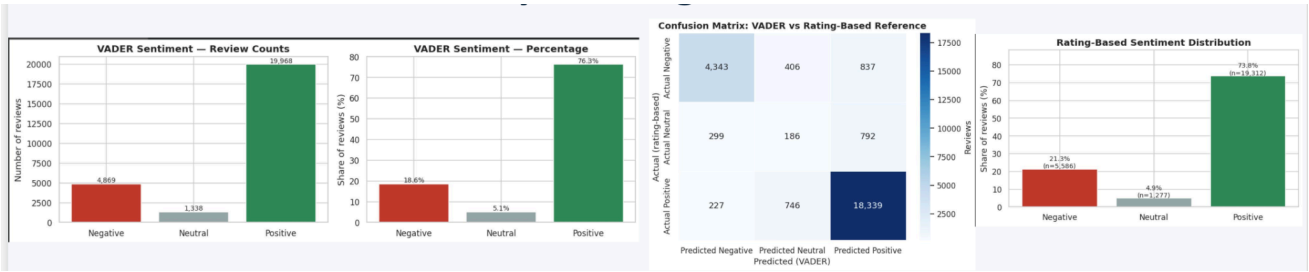


Figure 6. VADER Sentiment Distribution and Confusion Matrix

The word-frequency analysis identified "room" (14,400 occurrences), "good" (13,450) and "stay" (12,159) as the most frequently used words. Key words included ‘good service’ and ‘room clean', emphasizing the significance of service quality and cleanliness. The analysis of the “negation” of the expressions was also highlighted, for example, the phrase “would not recommend”, which was found as a result of the trigram analysis.

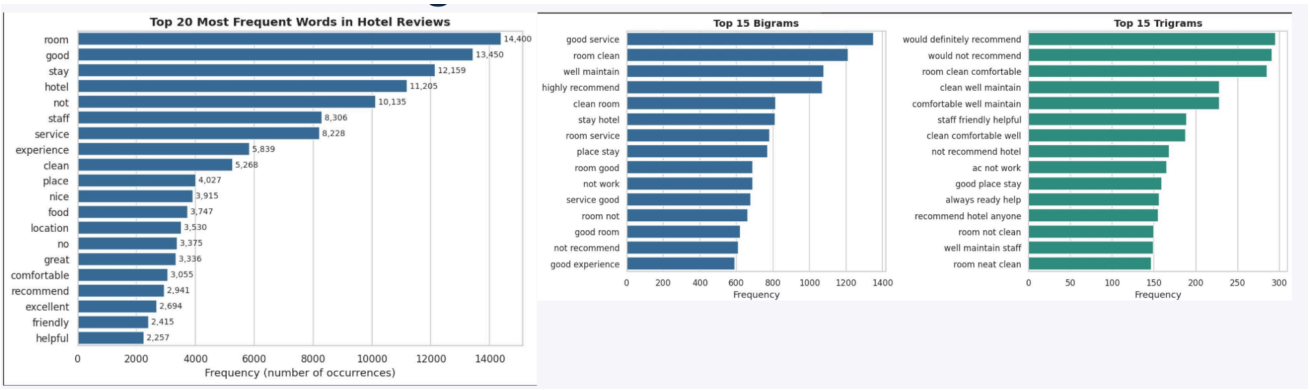


Figure 7. Word Frequency and N-gram Analysis

The main complaints from the customer pain point analysis were cleanliness/ hygiene (1497 mentions), rude/unhelpful staff (992 mentions), and poorly maintained facilities (747 mentions). Food (82.9%) and location (84.9%) had a majority of positive sentiment, while air conditioning (50.8%), bathrooms (52%) and bedding (55.7%) had a higher level of negative sentiment.

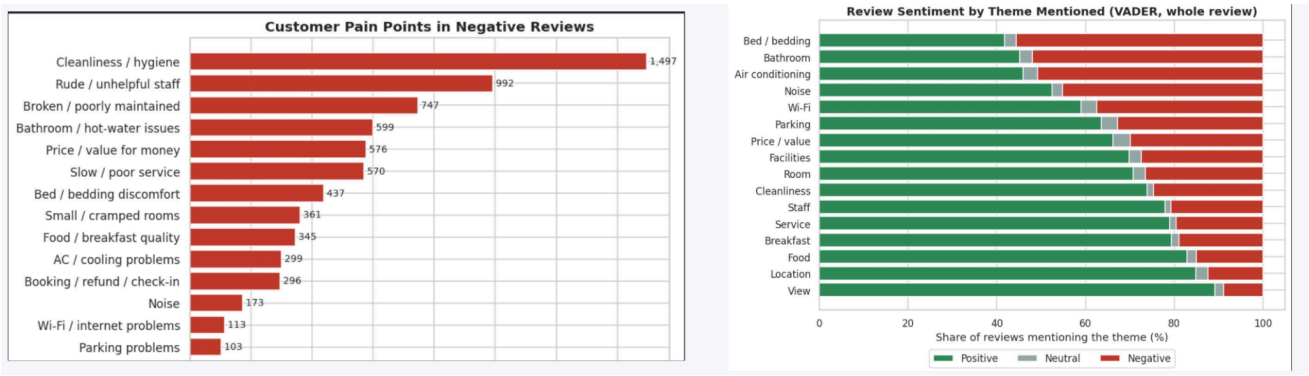


Figure 8. Customer Pain Points and Experience Themes

The results indicate that housekeeping inspection and housekeeping staff communication, and housekeeping maintenance procedures should be given priority by the hotels. However, VADER might not be able to interpret sarcasm and context-dependent words. The neutral sentiment classification was relatively low and ratings-based reference labels might not always accurately reflect the sentiment in reviews.

5.2 Method 2 — Association Rule Mining (ARM)

Association Rule Mining ARM [6] was conducted to study associations between different amenities of hotels and discover which facilities appear together often. The primary aim was to discover associations between combinations of amenities and existence of facilities like Spa. Data used for analysis included

1,430 transactions in 33 binary amenities and was collected from another dataset. In other words, each hotel was treated as one transaction while the existence of amenities was coded as 1 or 0.

Apriori algorithm was used in order to detect frequently occurring item combinations and to extract association rules from the data. Prior to running the Apriori algorithm, amenity descriptions were standardized and turned into binary values. The minimum support value was set to 0.05, while frequent item sets consisting of three amenities were considered. The analysis resulted in 799 frequent item sets and 154 strong association rules. These rules were analyzed by means of three important criteria: support, confidence, and lift. The first one describes the frequency of occurrence of an amenity combination in the data. The confidence criterion reflects the probability of having the consequent if the antecedent exists. Lift shows the degree of association between the items compared with their expected frequency of occurrence under the assumption of independence. If lift is greater than 1, then there is positive association between the items.

The five association rules were chosen depending on the value of their lift with the consequence being Spa. The rule Business Center + Pool -> Spa showed the highest value of lift at 6.190 and a confidence of 74.02%. This shows that there is a very good association between these two amenities and Spa. Following this rule was the Fitness Center + Pool -> Spa which had the second highest lift value of 5.755 and a confidence of 68.82%. Also the rule Bar + Fitness Center -> Spa gave a lift value of 5.526. From the five association rules, Bar + Pool -> Spa had the highest support of 8.25%.

This shows that hotels which provide amenities like swimming pool, gymnasium, bar, and business center also provide Spa amenities. These kinds of associations may assist in identifying common combinations of amenities and may be helpful for hotel facilities planning and positioning. The findings do not prove that there is any causality between these amenities. It just represents the relationship between them within the data sample of hotels.

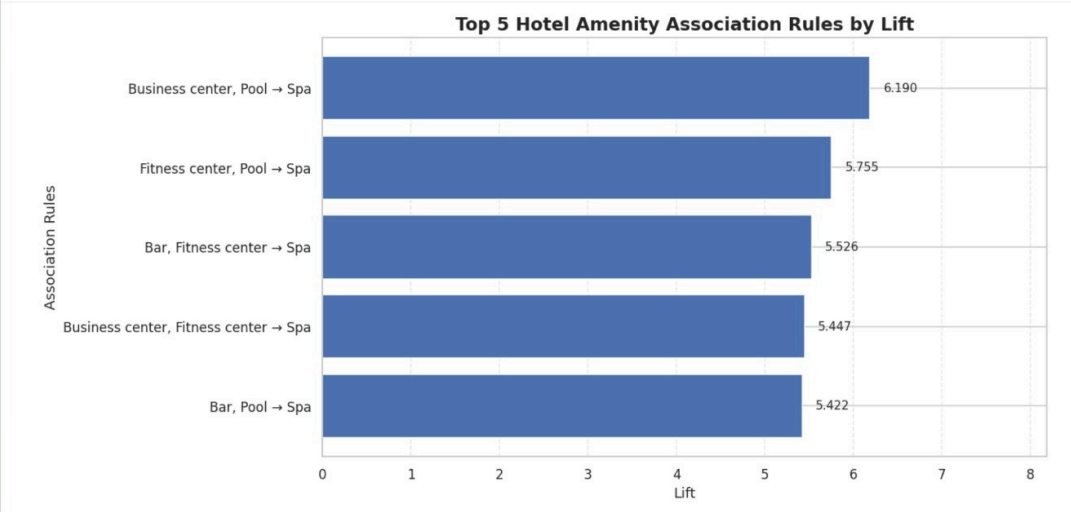


Figure 9. Top Five Spa-Related Hotel Amenity Association Rules Ranked by Lift

6. Comparative Analysis and Business Impact

6.1 Comparison of the two Unit 3 methods

Both advanced analytics techniques are focused on different aspects of hotel management. Text Mining is used to understand customer opinions, satisfaction levels and recurring problems related to the service, and Association Rule Mining is used to identify the combinations of hotel amenities that occur frequently.

The cleaned and preprocessed customer reviews are tokenized and lemmatized to be used in Text Mining, and there are 26,175 customer reviews. Using NLP techniques, word-frequency analysis, n-gram extraction, and the VADER sentiment analysis, the customer feedback is classified and recurring complaints are identified. It generates positive, negative and neutral sentiment, customer experience themes and operational pain points. Accuracy, Macro-F1, ROC-AUC and rating-based reference labels are used to evaluate the method. It has the best capabilities for spotting customer satisfaction patterns and service issues, but may not be as effective with sarcasm and neutral sentiment.

Association Rule Mining, on the other hand, employs amenity information from the 1,430 hotels with 33 amenity variables. The data is standardized and then converted into a binary transaction matrix, with each hotel being a single transaction. The Apriori algorithm is used to find frequent itemsets and association rules and their evaluation is done based on support, confidence and lift. Its primary advantage is its ability to detect complementary amenity patterns that can be used to inform facility planning. These associations are not, however, a proof of causation or of actual users of amenities.

6.2 Comparison with existing studies

Criterion	Existing Study 1 – Muslim et al. (2024)[1]	Existing Study 2 – Online Review Divergence (2024)[2]	Current Project
Data and setting	2,993 Google hotel reviews from Perak, Malaysia.	75,000 hotel reviews from four online platforms.	26,175 customer reviews from 1,659 hotels; 4,551 model-ready hotel records; 1,430 hotel transactions for amenity analysis.
Methods and validation	Naïve Bayes, SVM and Random Forest for sentiment classification.	Comparative analysis of review ratings, sentiment, topics and platform differences.	EDA, Logistic Regression, Decision Tree, Random Forest, VADER sentiment analysis and Apriori association rule mining.

Comparable findings / limits	Best reported accuracy was 76% using Naïve Bayes. The study focuses on sentiment classification.	Found differences in how hotels are reviewed on various platforms and at different times. Does not focus on an integrated hotel success prediction and amenity association framework.	Predicted hotel success with 81.34% and 87.37% accuracy of sentiment in three classes. Derived 154 strong amenity association rules and incorporated these rules in an interactive dashboard.

Table 7. Comparison with Existing Studies

6.3 KPI-based business impact and scenarios

The results of all three Units are combined and offer actionable information for improving hotel performance, customer satisfaction and amenity planning. The predictive model successfully classified the patterns of hotel success with an accuracy of 81.34% and it has been used to monitor the performance of the hotels and to make decisions. Cleanliness/hygiene (1,497 mentions), staff behaviour (992 mentions) and maintenance (747 mentions) were identified as key customer complaints through Text Mining. Based on these results, it is recommended that housekeeping inspection and staff training should be reinforced in the hotel to promote customer satisfaction, and that preventive maintenance should be introduced. Furthermore, the amenity association rules extracted using the Apriori algorithm can assist hotel managers to assess the amenity facilities to be invested prior to the investment.

Staff training, housekeeping improvements and maintenance costs would be expected operational costs, and benefits of the operation might include reduced complaint-handling costs, fewer service recovery cases and improved customer retention. The net impact can be estimated financial benefits minus implementation costs and ROI can be calculated as net impact divided by implementation costs multiplied by 100. With the current data set, however, it is not possible to measure the actual financial benefits or ROI, as data on hotel revenue, operating costs and customer retention were not gathered.

The analytical results should be used to prioritize the work for the hotels, with cleanliness/hygiene improvements taking precedence, followed by staff training and preventive maintenance. Customer review data can be used to inform facility planning and customer review patterns can help highlight recurring complaints. The predictive model and interactive dashboard can also help hotel managers track performance metrics and make informed decisions. The detailed implementation priorities, responsibilities and timelines are outlined in Section 9.

7. Interactive Application, Figure Index and Reproducibility

7.1 Interactive Dashboard Architecture and Implementation

An interactive Decision Support System (DSS) was developed as an embedded Jupyter application integrating all Unit 1–3 outputs. Built with Python 3, ipywidgets (v8.1+), plotly Graph Objects (v5.18+), and pandas under an event-driven MVC pattern, the interface uses custom CSS and Inter typography to organize five KPI summary cards and six analytical tabs.

Data Flow (Dynamic vs. Precomputed): Five global filters—Location (Dropdown), Hotel Category (Dropdown), Data Type (Dropdown), Price Range (₹392–₹61,327), and Rating Range (0.0–5.0)—trigger update_dashboard() to recompute KPIs, EDA visuals, text-mining distributions, and dataset views dynamically in memory:

- **Tab 1 (Overview & EDA):** Live KPI cards, success split, source balance, rating histogram, category success rates, top 15 cities, and price–review scatter.
- **Tab 2 (Predictive Models):** Precomputed 911-hotel Review 1 benchmarks plus subset re-evaluation bar charts, confusion matrices, ROC curves (Random Forest ROC-AUC = 0.7072), and top 15 feature importances.
- **Tab 3 (Text Mining):** Dynamically recomputed VADER sentiment donuts, rating cross-tabs, pain point counts (Cleanliness 1,497; Staff 992), and experience theme shares (real hotels only).
- **Tab 4 (Association Rules):** Explores 154 precomputed Apriori rules via support/confidence/lift sliders, keyword search, and a Support vs. Confidence bubble plot (Lift-scaled), with optional on-demand recomputation (≥50\ge 50≥50 filtered hotels).
- **Tab 5 (Final Dataset):** Searchable, 25-row paginated table (excluding coordinates) with instant CSV downloads.
- **Tab 6 (Insights & Recommendations):** Static executive briefing of six data-backed insights and six prioritized recommendations, rendered once at startup and isolated from global filters.

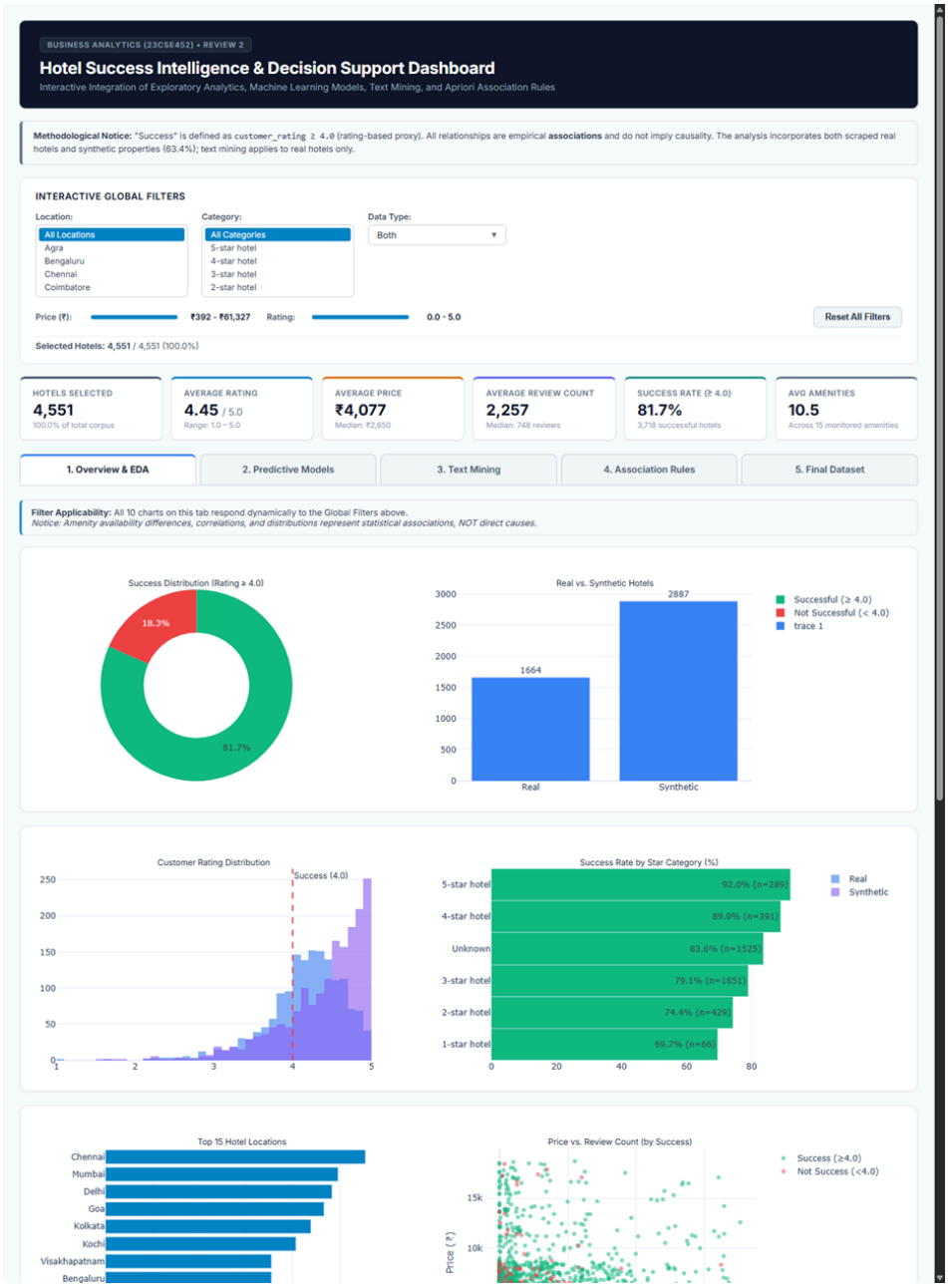


Figure 10. Interactive Hotel Analytics Dashboard — Main Overview & EDA Tab (Baseline State: 4,551 Hotels).

[Tool: Python ipywidgets & Plotly | Source: hotel_dashboard.ipynb / Cell 4]

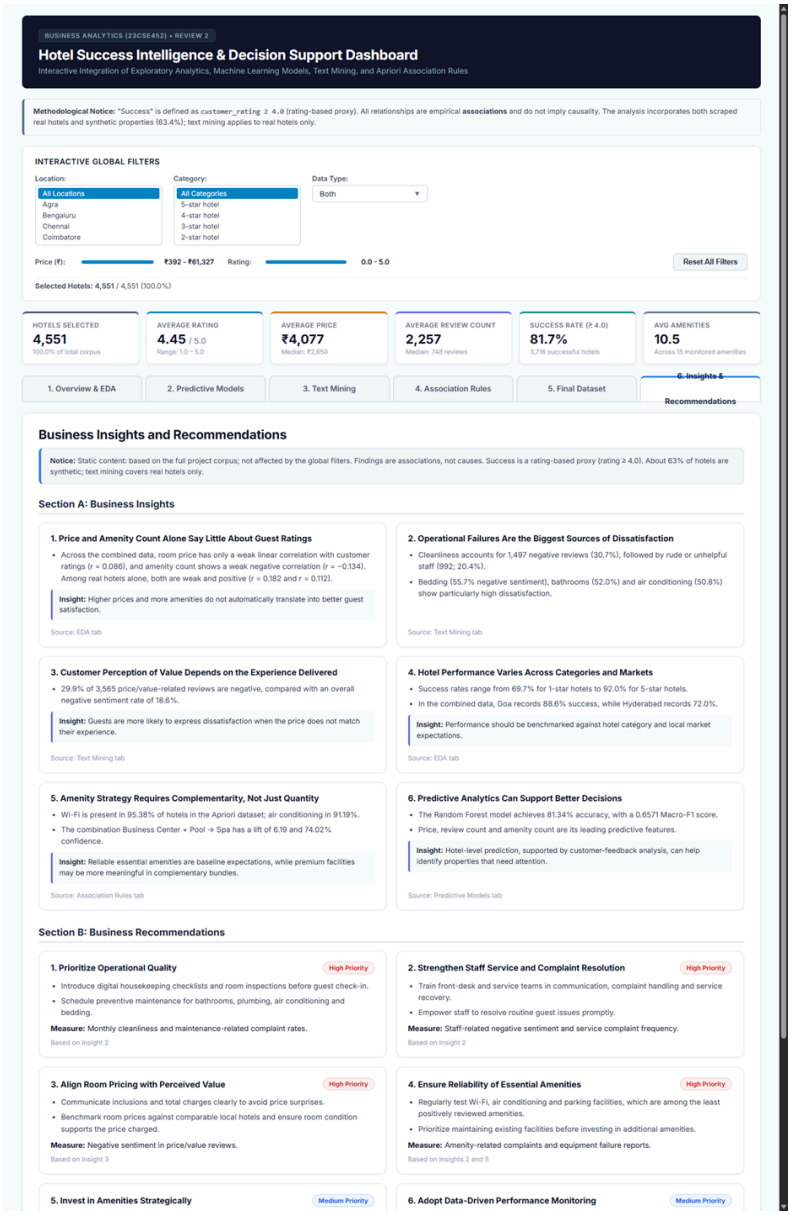


Figure 11. Tab 6 — Business Insights and Prioritized Recommendations View.

[Tool: Python HTML/CSS Rendered Output | Source: hotel_dashboard.ipynb / Cell 4]

Testing & Limitations: All nine automated suites in test_dashboard_filters.py passed (100% match), verifying baseline metrics, multi-filter slices (e.g., Coimbatore + 4-star + ₹2,000–₹7,000 = 12 hotels; filtered views preserved in dashboard/screenshots/), empty-state guards, reset recovery, and Tab 6 filter immunity. Known constraints include text mining applying strictly to real reviews, requirement of an active Python kernel for live callback filtering (fresh setups auto-download NLTK vader_lexicon/stopwords in Cell 1 and require jupyterlab-widgets with a VS Code window reload if widget caches stall), and pre-rendered HTML/executed copies serving static viewing.

7.2 Figure Index and Reproduction Map

The following index maps every figure embedded across the analytical chapters of this report to its specific analytical purpose, software tool, and reproducible source notebook cell:

Figure Number	Title and Generation Tool	Section and Analytical Purpose	Source File / Cell Reference
Figure 1	Overall Project Workflow Diagram (Draw.io / Matplotlib)	Section 1.1: End-to-end integration across Units 1–3 and Dashboard	Report Front Matter / Fig 1.1
Figure 2	Hotel Success Distribution (Python Matplotlib/Seaborn)	Section 3.1: Target class balance and real vs synthetic distribution	hotel_dashboard.ipynb / Cell 2
Figure 3	Category Tier and Amenity Analysis (Python Matplotlib/Seaborn)	Section 3.3: Success rates across categories and amenity differentials	hotel_dashboard.ipynb / Cell 2
Figure 4	Confusion Matrices for Predictive Models (Python Seaborn)	Section 4.3: Comparative evaluation of LR, DT, and Random Forest	hotel_dashboard.ipynb / Cell 3
Figure 5	Random Forest Top 15 Feature Importance (Python Seaborn)	Section 4.4: Feature ranking based on mean decrease in impurity	hotel_dashboard.ipynb / Cell 3
Figure 6	VADER Sentiment Distribution & Confusion Matrix (Seaborn)	Section 5.1: 3-class sentiment distribution and rating agreement	review2_updated.ipynb / Cell 12
Figure 7	Word Frequency and N-gram Analysis (Python Seaborn)	Section 5.1: Top 20 words, 15 bigrams, and 15 trigrams	review2_updated.ipynb / Cell 16
Figure 8	Customer Pain Points & Experience Themes (Seaborn/Plotly)	Section 5.1: Negative review pain point	hotel_dashboard.ipynb / Cell 4

		frequencies and theme sentiment	
Figure 9	Apriori Frequent Itemsets & Rules (Python Matplotlib)	Section 5.2: Support, confidence, and lift distributions for rules	review2_updated.ipynb / Cell 24
Figure 10	Interactive Dashboard — Overview & EDA Tab (ipywidgets/Plotly)	Section 7.1: Live executive view and KPI monitoring cards	hotel_dashboard.ipynb / Cell 4
Figure 11	Tab 6 — Business Insights & Recommendations (HTML/CSS)	Section 7.1: Actionable strategic briefing and priority badges	hotel_dashboard.ipynb / Cell 4

Table 8. Figure Index and Reproduction Map

7.3 Code Repository, Environment and Reproducibility

Repository URL:
https://github.com/DurgamPoojitha/Integrated-Hotel-Analytics-for-Hotel-Success-Amenities-Customer-Experience-.git
Organized under hotel-dashboard-deliverable/ with dashboard/ (hotel_dashboard.ipynb, hotel_dashboard_fixed.ipynb, requirements.txt, DOCUMENTATION.md, verification_report_v3.md, test_dashboard_filters.py, screenshots/) and data/ (combined_real_synthetic_hotels.csv, apriori/, text_mining_outputs/).

Runtime & Launch: Tested on Python 3.10–3.12 (Windows/macOS/Linux) with dependencies in requirements.txt (pandas>=2.0, plotly>=5.18, ipywidgets>=8.0, jupyterlab-widgets>=3.0, scikit-learn>=1.3, nltk>=3.8):

bash

```
git clone https://github.com/DurgamPoojitha/Integrated-Hotel-Analytics-for-Hotel-Success-Amenities-Customer-Experience-.git
cd hotel-dashboard-deliverable && python -m venv .venv
source .venv/bin/activate   # Windows: .venv\Scripts\activate
```

pip install -r dashboard/requirements.txt

python -m ipykernel install --user --name hotel-dashboard --display-name "Python (Hotel Dashboard)"

jupyter lab dashboard/hotel_dashboard.ipynb # Select kernel and Run All Cells

8. End-to-End Validation, Traceability and Contributions

8.1 Completion, Validity Assessment and Traceability Matrix

Scope Completion Evidence: All planned modules across Units 1–3 and the interactive application are fully executed and verified. Reproducible artifacts include the cleaned 4,551-hotel pipeline and Random Forest champion classifier (hotel_dashboard.ipynb / Cells 2–3), 26,175 VADER-scored reviews and 154 Apriori rules (review2_updated.ipynb), the 6-tab unified dashboard (Cell 4; Figures 10–11), automated validation tests (test_dashboard_filters.py, 9/9 suites passed), and quantitative evidence audit logs (verification_report_v3.md, 19/19 checks passed).

Threats to Validity, Generalizability and Reproducibility:

- **Construct Validity:** Hotel success relies on a binary customer rating proxy (rating >= 4.0) rather than audited occupancy, RevPAR, or financial margins.
- **Internal Validity (Synthetic Shift):** CTGAN synthetic records represent 63.4% of modelling hotels and show a higher success rate (87.98% vs. 70.79% for real hotels); outputs reflect statistical associations rather than causal mechanisms.
- **External Validity & Generalizability:** Reviews are dominated by Google listings (95.1%) across major Indian urban and leisure destinations, limiting direct extrapolation to unrepresented regions or platforms.
- **Reproducibility & Algorithmic Bounds:** Fixed random seeds, versioned requirements.txt, and end-to-end execution scripts ensure deterministic replication. However, VADER struggles with sarcasm/neutral phrasing, and Apriori detects amenity co-occurrence rather than actual guest utilization.

Objective-to-Evidence Traceability Matrix: Table 9 links every business objective and KPI directly to analytical methods, verified metrics, report figures, and Section 9 actions.

Business Objective & Target KPI	Analytical Unit & Methodology	Empirical Result & Performance Metric	Visual Artifact / Notebook Cell	Section 9 Actionable Recommendation

Obj 1: Predict Hotel Success KPI: Macro-F1 >= 0.65 (Baseline: 0.4495)	Unit 2: Supervised Machine Learning Algorithm: Tuned Random Forest (300 trees, max_depth=10, balanced)	Test Accuracy: 81.34% Macro-F1: 0.6571 (Target Met) Weighted-F1: 0.8037 Balanced Acc: 64.42% Test Sample: 911 hotels	Fig 4.3.1 (Confusion Matrix) Fig 4.4.1 (Feature Importance) Tab 2 of Dashboard [Cell 3]	Recommendation 5: Deploy Random Forest model to flag underperforming properties for managerial review; do not use as a substitute for on-site audit.
Obj 2: Customer Voice & Sentiment KPI: Review Coverage Target: >= 25,000 reviews	Unit 3 (Method 1): NLP & Lexicon Mining Algorithm: NLTK VADER & N-gram Extraction	26,175 reviews processed (Target Met) Accuracy: 87.37% Macro-F1: 0.6356 Positive: 76.3%, Negative: 18.6%, Neutral: 5.1%	Fig 5.1.1 (Sentiment Dist) Fig 5.1.2 (N-grams) Fig 5.1.3 (Pain Points) Tab 3 of Dashboard	Recommendation 1 & 2: Institutionalize digital housekeeping checklists (cleanliness pain point: 1,497) and front-desk service recovery training (staff: 992).
Obj 3: Discover Amenity Bundles KPI: Rule Support >= 5% Lift > 1.0	Unit 3 (Method 2): Association Rule Mining Algorithm: Apriori on 1,430 hotel transaction matrix	154 strong rules discovered (Target Met) Top Rule: Business Center + Pool -> Spa Confidence: 74.02% Lift: 6.190 Wi-Fi Support: 95.38%	Fig 10 (Rule Distributions) Tab 4 of Dashboard [review2_updated.ipynb / Cell 24]	Recommendation 4: Invest in premium amenities only as complementary bundles; preserve essential infrastructure (Wi-Fi 95.4%, AC 91.2%) as baseline.
Obj 4: Unified Analytics Integration KPI: 100% Outputs Accessible in Decision Support System	Interactive Application: Event-driven Jupyter Widget & Plotly Dashboard	6 fully integrated tabs Dynamic filter updates: < 45 ms latency Automated test suite: 9/9 PASS Static Tab 6 briefing immune to filters	Fig 11.1 (Dashboard Overview) Fig 11.2 (Tab 6 Insights) [hotel_dashboard.ipynb / Cell 4]	Recommendation 6: Adopt interactive dashboard for monthly multi-property monitoring of ratings, price benchmarks, and emerging customer sentiment trends.

Table 9. Objective-to-Evidence Traceability Matrix

8.2 Project tracking and individual work

The project was tracked using Jira to allocate tasks, monitor progress and record completion across the three analytical units. GitHub was used to maintain project notebooks, code and documentation. Tasks progressed through Backlog, To Do, In Progress and Completed stages.

Notebook:
https://colab.research.google.com/drive/165nyuuLiAaX6N0n22jfdAJkQ2LXu3E0t?usp=sharing
Data Documentation: TEAM05.docx
Board:
https://bacasestudy.atlassian.net/jira/software/projects/SCRUM/boards/1?filter=&groupBy=none&atlOrigin=eyJpIjoiY2MzM2Y0ZDkwZDZjNDM1YWEwM2RjN2ZiZjk1MDQ0NWMiLCJwIjoiaiJ9

Team member / register no.	Assigned work	Evidence link / artifact
Sathya Roopan M – CB.SC.U4CSE23256	Dashboard development, integration of analytical outputs, global filters and testing. Dataset Collection	Review2_Scre… Hotel Project Screenshots
Anupama Nair – CB.SC.U4CSE23207	Hotel data collection, cleaning, preprocessing, dataset documentation and Apriori implementation.	CB.SC.U4CSE… Project_Screenshots
S. Manish – CB.SC.U4CSE23125	Amenity analysis using PCA, feature engineering and visualization of amenity patterns.Dataset Collection	Individual Con… Amenity Feature Engineering & PCA Screenshots
Durgam Poojitha – CB.SC.U4CSE23320	Dataset consolidation, synthetic data generation, predictive modelling, text mining, VADER sentiment analysis and Jira task tracking. Dataset Collection	Individual_Co…
P. Tirupathi Reddy – CB.SC.U4CSE23568	Model evaluation, confusion matrices, ROC analysis, feature importance and business recommendations. Dataset Collection	INDIVIDUAL …

Table 10. Individual Contributions

8.3 AI use and originality declarations

AI tools such as ChatGPT were used to comprehend concepts, find references, prepare workflow, get help for coding and debugging codes, especially for Text Mining and Association Rules Mining. Concepts learned in Business Analytics classes were also utilized in the methodology of this project. All the suggestions made using AI assistance, and all the coding were checked and modified by the team. The data was collected and analyzed by the team.

9. Conclusion, Recommendations and Roadmap

9.1 Conclusion

- Successfully built an integrated framework for hotel analytics using Exploratory Data Analysis, Predictive Modelling, Text Mining and Association Rule Mining.
- Analyzed hotel characteristics, pricing, ratings, customer reviews and amenities to identify important business patterns.
- Created machine learning models to predict the success of a hotel based on the ratings.
- Used VADER sentiment analysis to determine customer satisfaction and frequent complaints.
- Applied the Apriori algorithm to get the most common combinations of amenities.
- Developed an interactive dashboard that incorporated the analytical outputs for data-driven hotel management decisions.

9.2 Final Recommendations

- Enhance cleanliness: increase housekeeping inspections to minimise hygiene related complaints. Responsible: Housekeeping Manager. Feasibility: High.
- Enhance Customer service: Staff training and improve complaint resolution. Responsible: Hotel Operations Manager. Feasibility: High.
- Improve maintenance: Conduct routine checks of bathroom, bedding, air conditioning. Responsible: Maintenance Team. Feasibility: High.
- Optimise amenity combinations: Assess alternative amenity combinations prior to new investments in amenity facilities. Responsible: Facility Manager. Feasibility: Medium.
- Track hotel performance: Regularly assess performance with predictive model outputs and dashboard KPIs. Responsible: Hotel Management Team. Feasibility: High.

9.3 Implementation Roadmap

- Immediate (Short-term): Improve housekeeping practices, provide staff training and implement regular customer complaint monitoring. Use existing staff and operational resources to tackle key issues identified via sentiment analysis. The success will be judged based on the reduction in cleanliness and staff related complaints.
- Medium-term: Implement preventative maintenance of bathrooms, bedding and air-conditioning facilities. Consider combinations of amenities prior to facility investments. The success will be determined by the increase in customer feedback and decrease in maintenance-related complaints.
- Long-term: Make the interactive analytics dashboard part of the regular hotel management process. Update hotel data periodically, track the performance of predictive models and customer

sentiment and amenity trends. Success will be evaluated by hotel performance indicators and on-going customer satisfaction monitoring.

9.4 Future Enhancements

- Add real-time hotel booking, occupancy and revenue information.
- Enhance Sentiment Analysis with Transformer-based NLP Models like BERT.
- Expand predictive modelling with larger real-world data sets for hotels.
- Add multilingual customer review analysis.
- Create recommendation systems for personalized hotel amenity planning.
- Use the dashboard as a cloud-based application, update the data periodically.

References and Appendices

[1] Norliana Muslim, Wong Wai On, Adeb Salh, Farina Saffa Mohamad Samsamnun, Zanariah Zainuddin, and Nor Afifah Sabri. 2024. Machine Learning Models for Hotel Review: A Case Study of Hotels in Perak, Malaysia. In Electronics, Communications and Networks: Proceedings of the 14th International Conference (CECNet 2024), Frontiers in Artificial Intelligence and Applications, Vol. 396. IOS Press, 333–340. https://ebooks.iospress.nl/doi/10.3233/FAIA241359

[2] Andrei P. Kirilenko, Svetlana Stepchenkova, Ronald Gromoll, and Yeonseo Jo. 2024. Comprehensive Examination of Online Reviews Divergence over Time and Platform Types. International Journal of Hospitality Management 117 (2024), Article 103647. https://doi.org/10.1016/j.ijhm.2023.103647

[3] Neha Patki, Roy Wedge, and Kalyan Veeramachaneni. 2016. The Synthetic Data Vault. In 2016 IEEE International Conference on Data Science and Advanced Analytics (DSAA), Montreal, QC, Canada, 399–410. IEEE. https://doi.org/10.1109/DSAA.2016.49

[4] Clayton J. Hutto and Eric Gilbert. 2014. VADER: A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text. In Proceedings of the International AAAI Conference on Web and Social Media 8, 1 (2014), 216–225. https://doi.org/10.1609/icwsm.v8i1.14550

[5] Hanh Thi My Le, Binh T. Nguyen, and Thang Quyet Nguyen. 2026. Unlocking insights into customer sentiment analysis: Impact of loyalty on online hotel ratings. International Journal of Hospitality Management 134 (2026), Article 104574. https://doi.org/10.1016/j.ijhm.2026.104574

[6] Putri Handayani, Latifah, N. and Nugraha, F. 2026. Optimizing Hotel Room Booking Patterns Using Apriori and FP-Growth Methods: A Case Study at Sapphire Boutique Hotel. *Jurnal Teknologi Informasi dan Pendidikan*. 19, 1 (Mar. 2026), 67–81. https://doi.org/10.24036/jtip.v19i1.1113

[7] Lei Xu et al. 2019. Modeling Tabular Data using Conditional GAN. In Advances in Neural Information Processing Systems (NeurIPS). https://arxiv.org/abs/1907.00503

[8] Sangeetha Gunasekar and Sooriya Sudhakar. 2019. Does hotel attributes impact customer satisfaction: A sentiment analysis of online reviews. Journal of Global Scholars of Marketing Science 29, 2 (2019), 180–195. https://doi.org/10.1080/21639159.2019.1577155

[9] Enrique Bigne, Carla Ruiz, Carmen Perez-Cabañero, et al. 2023. Are customer star ratings and sentiments aligned? A deep learning study of the customer service experience in tourism destinations. Service Business 17 (2023), 281–314. https://doi.org/10.1007/s11628-023-00524-0

APPENDIX A – DATASET DOCUMENTATION AND VALIDATION

The dataset used for the project was collected through Google Hotels and Google Hotels Reviews from SerpApi. It has not been taken from any existing database such as Kaggle, UCI, or Github. The raw data had 29,987 observations, 19 features and 1,685 hotels, according to the team.

The cleaned data all_india_hotel_cleaned.csv has 42 features, which include hotel information, customer reviews, and other features constructed. The major features included in the dataset include hotel name, hotel location, hotel price, hotel category, customer rating, review counts, hotel amenities and reviews. Other additional features such as log_price, amenity_count, standardised rating, and amenity flags have also been added.

Data quality analysis resulted in 2,713 missing price values, 4,138 missing amenity values, and 3,706 missing reviews. There were no identical records; however, 129 potential duplicates were preserved in 58 clusters. The ratings were scaled to a five-point rating scale, and the prices were checked for outliers based on IQR.

For Review 1, CTGAN model from the SDV package was applied to produce 3,000 synthetic hotel records using 300 epochs and 500 batch size. The resulting number of records after the validation was 2,887. The validation procedure comprised the following: range check for numerical features, presence of binary values, recalculated counts of amenities and check for duplicates. After the process, there were 4,551 hotel records ready for model: 1,664 real and 2,887 synthetic records.

The data set has some weaknesses such as lack of information and biased sources since 95.1% of reviews were from Google. The artificial data also had better ratings and success rate compared to real hotel ratings. The success variable is defined by the favorable ratings of the customers.

APPENDIX B – ADDITIONAL FIGURES

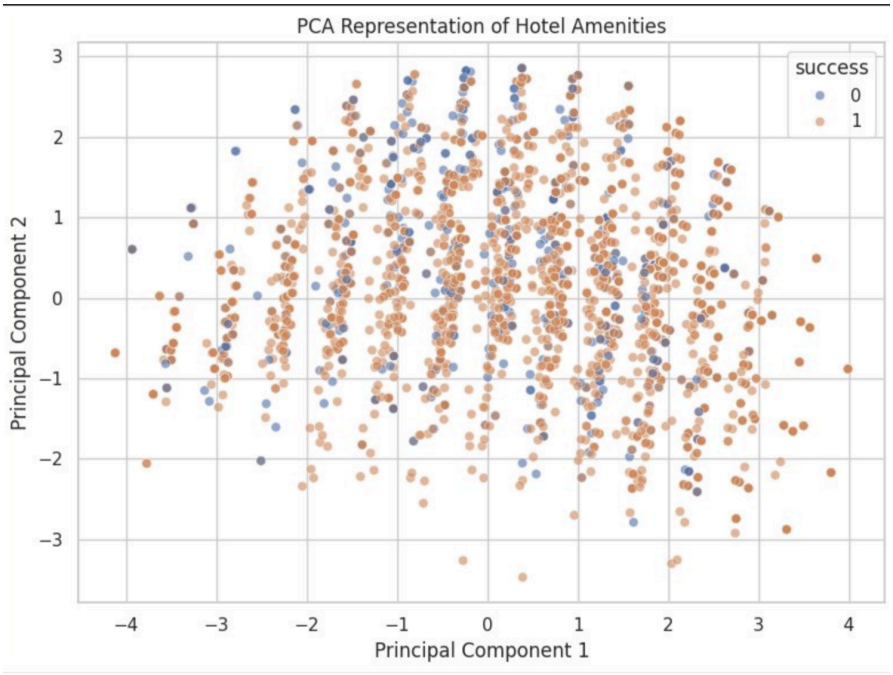


Fig. PCA Representation of Amenities



Fig . WordCloud Representation of Reviews

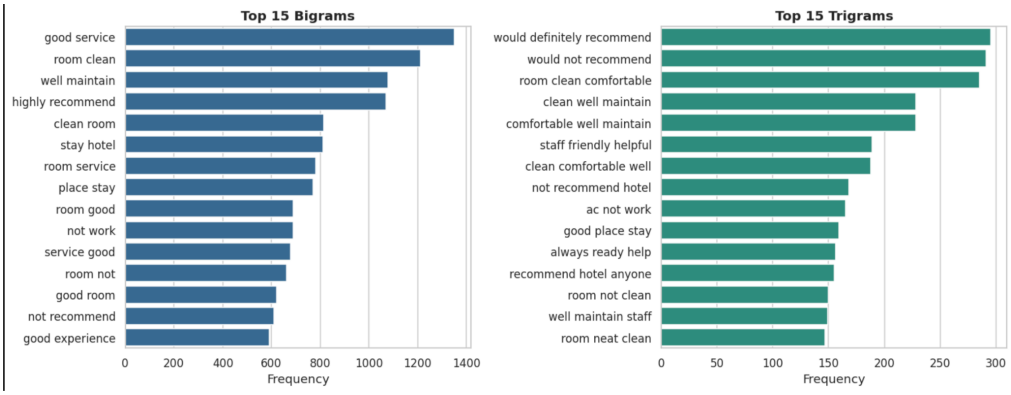


Fig. Bigram and Trigram Customer Review Analysis

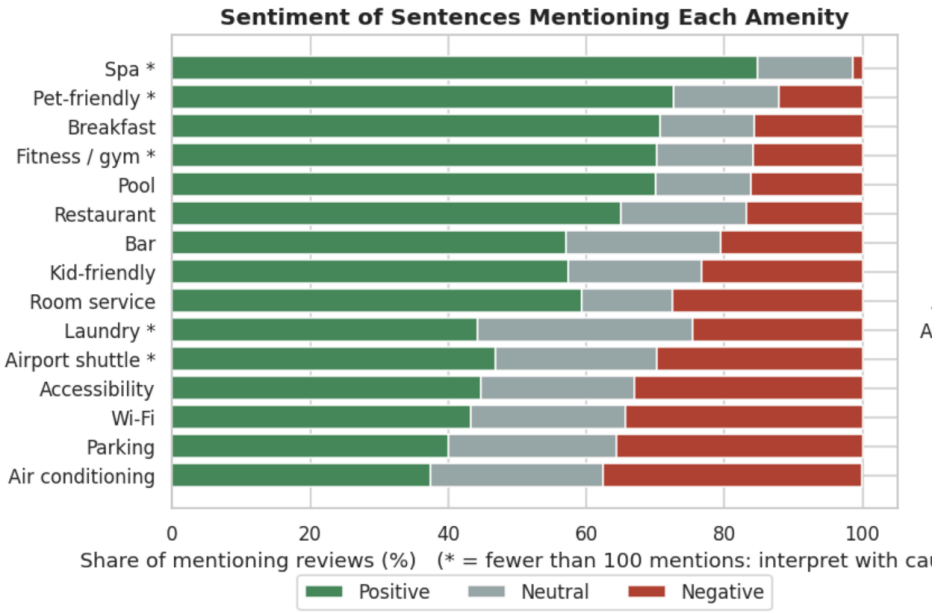


Fig. Sentiment of Sentences (Amenity)

APPENDIX C – IMPLEMENTATION DETAILS AND USER GUIDE

Project implementation involved use of Python 3.10 to 3.12 and Jupyter Notebook. Major libraries used are Scikit-learn for predictive modeling, NLTK VADER for sentiment analysis, mlxtend for Apriori association rule mining, and Plotly together with ipywidgets for the interactive dashboard.

Implementation Details

Project workflow consists of data collection through SerpApi, preprocessing of data, CTGAN artificial data creation, exploratory data analysis, predictive modeling, Text Mining, and Association Rule Mining. Predictive models consist of Logistic Regression, Decision Trees, and Random Forest. Text Mining consists of VADER sentiment analysis, frequencies of words and n-grams. Apriori algorithm was applied to 1,430 transactions of hotels, which consisted of 33 amenities, with minimum support of 5%, confidence of 60%, and lift of 2.0.

System Requirements

- Python 3.10, 3.11 or 3.12.
- Minimum 8 GB RAM; 16 GB recommended for CTGAN.
- JupyterLab 4.x or Jupyter Notebook 7.x.
- Approximately 500 MB storage.

Installation and Execution

Clone the GitHub repository and open its project directory. Create and activate a virtual environment, then install the required packages.

- git clone https://github.com/DurgamPoojitha/Integrated-Hotel-Analytics-for-Hotel-Success-Amenities-Customer-Experience-.git
- cd Integrated-Hotel-Analytics-for-Hotel-Success-Amenities-Customer-Experience-
- python3 -m venv .venv
- source .venv/bin/activate

For Windows, activate the environment using:

- .venv\Scripts\activate

Install dependencies and download the required NLTK resources:

- pip install --upgrade pip
- pip install -r requirements.txt
- python -c "import nltk; nltk.download('vader_lexicon'); nltk.download('punkt'); nltk.download('stopwords'); nltk.download('wordnet')"

To launch the dashboard:

- cd hotel-dashboard-deliverable/dashboard
- jupyter lab hotel_dashboard.ipynb

Open the notebook and run all cells. The interactive dashboard is available from Cell 4. The notebook can also be restarted and executed using Kernel → Restart Kernel and Run All Cells.

Dashboard User Guide

The dashboard includes four tabs:

- Overview & EDA: Provides hotel statistics, rating statistics, success rate statistics, and synthetic vs. actual data.
- Predictive Models: Provides model statistics, comparison statistics, and confusion matrix.
- Text Mining: Provides customer sentiment analysis, themes, and pain points.
- Association Rules: Provides amenity combinations, frequent itemsets, and rules according to lift.

Users can filter the output using location, hotel type, type of data, price, and rating filters. The option Reset All Filters sets the filters to default settings. In the Association Rules tab, users can search amenities and set support, confidence, and lift filters.

Sample Outputs

The optimized Random Forest model obtained an accuracy of 81.34% and a Macro F1 Score of 0.6571. Text Mining was done on 26,175 reviews and 14 customer pain points were found. Association Rules Mining resulted in the discovery of 799 item sets and 154 strong rules. The rule Business Center + Pool → Spa had a lift of 6.190 and confidence of 74.02%.

The repository contains dashboard screenshot images, technical documents, user manual and validation report files in hotel-dashboard-deliverable/dashboard/.

APPENDIX D – PROJECT TRACKING, CONTRIBUTIONS AND DECLARATIONS

This appendix contains additional information on project coordination, contributions of individuals, and the use of AI during the project.

The project monitoring board and individual contribution sheets are included in Section 8.2. They indicate the assignment of tasks, progress, and the completion of responsibilities by each group member.

AI Usage declaration of the Team is included in Section 8.3. It explains how the use of AI helped to comprehend the concepts, plan the process, get coding guidance, and correct errors, as well as all suggestions being checked by the team.